Fiscal Year:	FY 2014	**Task Last Updated:**	FY 01/08/2014
PI Name:	Berkowitz, Dan E. M.D.		
Project Title:	Radiation Induced Cardiovascular Oxidative Stress: Thiol Modification of Proteins and Novel and Sensitive Biomarkers		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	21287-8711	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2012	**End Date:**	10/01/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	McComb, Mark (Boston University School of Medicine)		
Grant/Contract No.:	NCC 9-58-CA00002 ; Directed Research		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We have used a tiered approach in this pilot project to detect biomarkers in plasma associated with exposure to radiation as well as altered redox biology and oxidative post translational modifications (OPTMs) on tissue resident proteins from animals exposed to radiation. We proposed a series of studies in mice to measure protein and OPTM change occurring in the heart and vasculature with subsequent detection in plasma. Proteomic studies in tissues of these mice will enable construction of databases of tissue-resident proteins with specific OPTM on modified peptides. In proteomic studies of the plasma of the mice, we will be able to cross-reference the OPTM-modified peptides found in the tissue studies, developing a database of tissue-resident proteins with OPTM that are found in the blood. This database will be used to search human patient plasma databases of OPTM in two ways. First, homologous human sequences of mouse proteins with OPTM will be sought in human plasma. Second, we will affinity purify and identify specific OPTM (e.g., nitrotyrosine, lipid) in human plasma, and cross reference them to the mouse tissue protein database. Finally, OPTM on circulating plasma proteins that are not contained in the tissue database (e.g., albumin, transthyretin, etc.), will be will be		

	evaluated as potential markers. To this point, we have identified protein/peptide and OPTM changes that are dependent upon dose and reflective of CVD. PTMs yield both specific & non-specific changes across the sample groups. The results in this mouse model correlate well with our other mouse models of cardiovascular disease (CVD) and our human models of CVD. Information gained from these CVD models are applicable to other oxidative stress-induced models and applicable to other cardiovascular diseases. Discovery of potential biomarkers: peptide, proteins, OPTMs will allow us to gain insight into the etiology of radiation induced CVD, understand host physiological response to pathogenesis and afford earlier detection, improve diagnosis and potential treatment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Development of a CVD-specific protein / OPTM and a space radiation-induced/ OPTM panel obtained from mouse and patient models will afford the first step in biomarker panel development such that radiation-induced cardiovascular disease and progression may be performed directly at the molecular level. Correlation of proteomics data with phenotype and genomics data will lead to a better understanding of the effects of radiation and the host physiological response to the pathogenesis of radiation-induced CVD. The discovery of potential biomarkers will afford earlier detection, improved diagnosis and interventional strategies not only to reduce CVD risk associated with space flight but other cardiovascular diseases in which oxidative stress is a significant etiological component.
Task Progress:	We have used a tiered approach in this pilot project to detect biomarkers in plasma associated with exposure to radiation as well as altered redox biology and oxidative post translational modifications (OPTMs) on tissue resident proteins from animals exposed to radiation.
Bibliography Type:	Description: (Last Updated: 01/13/2014)